

PRS821S PROGRAMMING FOR SECURITY PROFESSIONAL, SEM 1, 2026

WEEK 5C LECTURE NOTES

FILE AND MEMORY MANAGEMENT SECURITY

File and memory management security ensures system integrity by controlling access to data and resources. It prevents unauthorised processes from reading or modifying protected memory regions, manages file permissions (read, write, execute), and isolates applications, thereby protecting against malware and data corruption through techniques such as paging and segmentation.

- **Memory Management Security:** Focuses on protecting data stored in RAM. It ensures that one process cannot access the memory allocated to another, preventing data breaches or corruption. Techniques include memory protection (segmentation/paging), address space layout randomisation (ASLR), and preventing buffer overflows.
- **File Management Security:** Controls access to files and directories stored on persistent storage. It involves implementing access control lists (ACLs) to determine user permissions (read, write, execute), file encryption, auditing file access, and ensuring data integrity.
- **Key Security Requirements:** Memory management ensures protection, sharing, and logical/physical organisation.
- **Operating System Role:** The OS keeps track of which parts of memory are currently being used, by whom, and handles security by allowing only authorised access to resources.
- **Protection against Threats:** These systems guard against malware, data theft, and denial-of-service attacks by ensuring that authorised programs have unrestricted access, while unauthorised programs are blocked.

Key Security Mechanisms:

- **Memory Protection:** Prevents unauthorised memory access, resulting in faults (e.g., segmentation fault) to stop suspicious processes.
- **Access Control:** Restricts file and system access based on user credentials.
- **File Organisation:** Uses structured approaches (directories, folders) and metadata to maintain file integrity and security.
- **Virtual Memory:** Allows for efficient, isolated, and protected memory usage across multiple processes.

MEMORY MANAGEMENT SECURITY protects RAM data integrity and confidentiality by isolating processes, preventing unauthorised access, and defending against exploits like buffer overflows. It uses hardware/software mechanisms such as memory protection (paging/segmentation), Address Space Layout Randomisation (ASLR), and secure coding to ensure that programs access only their assigned memory.

Key Aspects of Memory Security:

- **Memory Protection:** A fundamental security feature that prevents a faulty or malicious application from crashing the system or accessing data belonging to other applications or the operating system.
- **Isolation & Access Control:** Ensures that each process operates within its own designated memory space, preventing one process from reading or writing to another's memory.
- **Virtual Memory:** Uses virtual address spaces to map to physical memory, ensuring processes cannot directly access unauthorised physical RAM.
- **Techniques & Mitigations:**
 - **ASLR (Address Space Layout Randomisation):** Randomises memory addresses used by system/applications to hinder attacks.
 - **Stack Canaries:** Detects stack buffer overflows before malicious code executes.
 - **Non-Executable Pages:** Prevents code execution in data-only memory regions.
 - **Memory-Safe Languages:** Using languages like Rust, Go, or Java can inherently prevent common memory errors.
- **Key Requirements:** Secure memory management must provide relocation, protection, sharing, and logical/physical organisation.
- **Hardware Role:** Memory Management Units (MMUs) enforce these restrictions by checking permissions before accessing memory.

FILE MANAGEMENT SECURITY involves protecting digital files from unauthorised access, alteration, deletion, and theft through techniques like encryption, access controls, and monitoring. It ensures data confidentiality, integrity, and availability by managing permissions—who can read, write, or execute files—and implementing audit trails to detect suspicious activity.

Key Aspects of File Management Security:

- **Access Control:** Limits user access to specific files and directories, ensuring only authorised individuals can view or edit.
- **Encryption:** Protects sensitive data at rest and in transit, rendering it unreadable to unauthorised users.
- **Audit Trails & Monitoring:** Tracks and logs file access to detect, record, and respond to potential security breaches.
- **Data Integrity & Backup:** Prevents file tampering and ensures data recoverability through regular backups and version control.
- **File Structure Protection:** Organises files securely to prevent unauthorised access, particularly by avoiding storing critical data in system directories.

Common Threats and Protections:

- **Threats:** Unauthorised access, data theft, tampering, ransomware, and accidental deletion.
- **Protections:** Implementing strong, granular permissions, using encryption for storage, conducting regular audits, and maintaining secure backups